

YugaByteDB: A Distributed Cloud Native Database for a Highly Scalable Data Store

This article introduces YugaByteDB, an open source, cloud native relational DB for powering global, Internet scale apps. YugaByteDB has the features of both SQL as well as NoSQL databases, giving rise to a hybrid database system that is cloud native.

First generation database systems are SQL based (DBMS, RDBMS) and are suitable for relational data models. The second generation database systems are NoSQL (Not only SQL) based (MongoDB, Hadoop), and are suitable for semi-structured or unstructured data models. There are many situations in which first generation database systems miss some of the functionalities of second generation DB systems and vice versa. To address this gap, the current trend is towards third generation NewSQL database systems.

NewSQL DB systems have most of the features of SQL as well as NoSQL, and work as hybrid database systems. A trending NewSQL DB system is YugaByteDB, an open source distributed database that is cloud native and suitable for highly scalable data access with geo-distributed access.

In general, SQL is more convenient for fast writes to databases and NoSQL is for fast reads from a database, whereas NewSQL DBs like YugaByteDB are suitable for both fast read and write since they are cloud native solutions. It is most convenient to use them for large enterprise solutions such as data lakes, batch processing and live visualisation reports from massive data processing.

YugaByteDB is not only distributed and cloud native but also ensures high performance, Internet-scale and geo-



distributed service. The architecture and core layers make it cloud-agnostic and available for on-premise deployments (hybrid cloud). It fully supports distributed and container deployments. YugaByteDB is designed for extensibility with the objective of re-using PostgreSQL's query layer code in its internal layering architecture.

It supports distributed transactions for large scale processing. YugaByteDB is a consistent and partition tolerant (CP tolerant) database and hence achieves very high availability. Its architectural design is very similar to that of Google Cloud Spanner, which is also a CP system, by design. The design of YugaByteDB ensures that the complete database cluster (irrespective of the single or multiple nodes in its deployment instance) makes it available to applications as a single logical SQL database like any other common RDBMS. Hence, it comes with the flexibility of PostgreSQL and the easy-to-understand ACID (atomicity, consistency, isolation and durability)

transaction semantics for applications that make it easy to use.

Starting off with YugaByteDB

YugaByteDB is available as a local installation (to install in Windows, Linux, etc, as a standalone DB). The YugaByte platform, as a multi-node clustered installation, is suitable for the private or public cloud (SaaS) in AWS, Azure, GCP or other pivotal cloud service provider environments and also as YugaByte Cloud, a PaaS platform, as a containerised solution (Dockerized).

YugaByteDB has an API interfacing service to run SQL commands through YSQL (YugaByte SQL) and can use standard ANSI commands as used in PostgreSQL. This is because YugaByteDB is based on a PostgreSQL query layer as a client interface.

Figure 1 shows how to connect to a single-node YugaByteDB instance and run YSQL commands to create a database and table, and also to insert records to the table.

An application built for a PostgreSQL database with the best one-node performance can scale out without changing the application when moving to YugaByteDB, and vice versa.

An overview of YSQL

YugaByteDB is an open source database and it can run anywhere — on multi-cloud environments, the hybrid cloud and also in a containerised Kubernetes environment. At the same time, it

